Heliyon 9 (2023) e14545

Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon

Research article

Exploring the essence of compound fault diagnosis: A novel multi-label domain adaptation method and its application to bearings

Liuxing Chu [a], Qi Li [a], Bingru Yang [a], Liang Chen [a,*], Changqing Shen [a], Dong Wang [b]

[a] School of Mechanical and Electric Engineering, Soochow University, Suzhou, 215000, China
[b] The State Key Laboratory of Mechanical System and Vibration, Shanghai Jiao Tong University, Shanghai, 200240, China

ARTICLE INFO

Keywords:
Compound fault diagnosis
Domain adaptation
Multi-label learning
Rolling bearing

ABSTRACT

Compound fault diagnosis in essence is a fundamental but difficult problem to be solved. The separation and extraction of compound fault features remain great challenges in industrial applications due to the lack of labeled fault data. This paper proposes a novel multi-label domain adaptation method applicable to compound fault diagnosis of bearings. Firstly, multi-layer domain adaptation is designed based on a fault feature extractor with customized residual blocks. In that way, features from discrepant domain can be transformed into domain-invariant features. Furthermore, a multi-label classifier is applied to decompose compound fault features into corresponding single fault feature, and diagnoses them separately. The application on bearing datasets demonstrates that the proposed method could enhance the detachable degree of compound faults and achieve greater diagnostic performance than other existing methods.

1. Introduction

Data-driven fault diagnosis technologies have attracted widespread attention recently. For rolling bearings fault diagnosis, deep learning techniques could extract the fault features intelligently without expert knowledge, thus being widely applied. The existing researches are mostly aimed at the diagnosis of single fault. However, in practical application, bearing faults may frequently appear in the form of compound faults composed of multiple single faults [1]. Compound faults typically take place when multiple critical parts of machinery are damaged at the same time, and they may result in catastrophic accidents. Compared with the single fault, compound faults are more harmful and more difficult to be identified [2]. Therefore, the research on compound fault diagnosis is both significant and challenging. The main problems for compound fault diagnosis are listed below.

1) The vibration signals with compound faults usually contain strong noise and nonlinearly coupled features interfered with each other [3]. Therefore, it is difficult to accurately decouple them in theory, tending to cause misidentification in the diagnosis.
2) The compound fault signals have no clear periodicity, and their amplitude changes greatly. The peak value at the fault characteristic frequency is not obvious, and the weak fault signal is almost obscured [4]. Traditional signal processing methods in time and frequency domain have trouble in extracting the fault features of compound faults.

* Corresponding author.
E-mail address: chenl@suda.edu.cn (L. Chen).

https://doi.org/10.1016/j.heliyon.2023.e14545
Received 1 October 2022; Received in revised form 5 March 2023; Accepted 9 March 2023
Available online 11 March 2023

3) Existing methods treat the compound faults as a new kind of fault pattern, independently of the single fault, while ignoring the potential relationship between them [5]. Actually, these methods are pseudo-diagnosis for compound faults, and unable to capture the essential features.
4) Moreover, in practical application, the data from target task often has different distributions with from training data [6], which will further degrade the model's performance. Existing deep learning-based methods depend on plenty of labeled training data. However, it is laborious even impractical to collect and mark the labels for the compound fault data in the target domain, which is the biggest restriction for engineering applications of data-driven fault diagnosis method.

Therefore, the research on compound fault diagnosis is of great significance, especially for industrial applications lacking of labeled fault data.

In view of the first and second problems mentioned above, i.e., the compound fault signal contains complex coupling features, which makes the traditional signal processing methods powerless. There have been some progresses in literature. Wang et al. [7] presented a new approach based on ensemble empirical mode decomposition (EEMD) to decompose compound fault features. And independent component analysis (ICA) technology is applied to make the fault features more easily identified. Yan et al. [4] made use of optimal variational mode decomposition to separate the compound signals and the inherent modal components containing the main fault features are adaptively selected. Luo et al. [8] used an extreme learning machine (ELM) to obtain compound fault features rapidly. And gravitational search algorithm (GSA) is used to seek essential features during iterations.

The above advanced signal processing methods, however, requires expert knowledge of each type of compound fault feature. And only when the signal source satisfies a series of statistical assumptions can the compound signal be decoupled effectively. Besides, these methods need to extract features manually, which relies on professional knowledge and a complicated tuning process. On the other hand, the diagnosis models based on the above methods usually apply only to specific problems and conditions, with low flexibility to adapt to the variable working conditions. As we know, the difficulty and time required for modeling will increase exponentially as the complexity and functionality of the equipment increase [9]. Therefore, in order to improve the intelligence and robustness of the compound fault diagnosis, data-driven method is the proper approach for liberating the diagnosis from dependence on expert experience and knowledge.

In view of the third problem mentioned above, i.e., compound faults are difficult to be decoupled as the single faults. Multi-label learning has an important advantage for this problem because it can solve the classification problem that one instance is associated with more than one class labels. Multi-label learning has been wildly used to a variety of problems, including text classification, image annotation, gene function analysis, and other issues. Sun et al. [10] applied multi-label learning to image categorization, and use latent factors to build dependency structure. Adding a sparseness prior to the association of labels and factors, thus addressing the problem that the model over-considered label correlations. Yang et al. [11] proposed to view the multi-label classification task in natural language processing as a sequence generation problem to model the correlations between labels. And a sequence generation model with a novel decoder structure is proposed to improve the performance of classification. Zhu et al. [12] were concerned about label correlations in multi-label learning and exploited global and local label correlations simultaneously, through learning a latent label representation and optimizing label manifolds. For fault diagnosis, Vong et al. [13] proposed a pairwise comparison of multi-label classification to diagnose multiple faults that occur simultaneously in the ignition system of an automobile engine. Georgoulas et al. [14] applied the problem transformation strategy in multi-label learning to appropriately represent the faults that occurred concurrently in an induction motor, and combined with the classifier to achieve an efficient diagnosis. Yu et al. [15] came up with a multi-label convolutional neural network incorporating meta-learning to address few-shot fault diagnosis.

Inspired by the above works, this paper introduces the multi-label learning to encode and classify the bearing compound faults. And a deep residual convolution neural network is combined to construct an adaptive fault diagnosis model. The combination of multi-label learning and deep neural network not only gets rid of the dependence of expert knowledge required by traditional signal processing methods, but also overcomes the weakness of deep learning methods in ignoring the relationship between compound faults and single fault. Hence, the compound fault diagnosis in this way is more rigor and reliable.

In view of the fourth problem mentioned above, i.e., the labeled compound fault data is always unavailable on target domain. Transfer learning maybe a powerful solution for this issue. Tong et al. [16] adopted a transfer learning method under variable operating conditions, which closes the divergence between training data and test data, thereby a transferable feature representation is achieved. Han et al. [17] proposed a novel fault diagnosis framework, which can exploit the discrimination structures associated with the labeled data to adapt the conditional distribution of unlabeled target data. Zhong et al. [18] developed a transfer learning method based on CNN and SVM. And a feature mapping model was used to compute the feature representations to solve the problem of gas turbine fault diagnosis with small samples. Lu et al. [19] mapped the data into a Reproducing Kernel Hilbert Space (RKHS), and used a weight regularization term to enhance the representative features of the original data. Wu et al. [20] constructed a long-short term memory recurrent neural network model based on instance-transfer learning to generate auxiliary datasets. Meanwhile, transfer learning is utilized to reduce the discrepancy between feature distributions. Our research team also proposed a novel method based on knowledge mapping to explore domain-invariant knowledge by adversarial training [21], and set up an adversarial domain generalization fault diagnosis framework [22]. Through the game between the generator (feature extractor) and the domain discriminator, the domain-invariant feature could be successfully extracted.

In general, this paper proposes a novel deep domain adaptation network with multi-label learning, which can tackle the above problems and is applicable to compound fault diagnosis of industrial bearings. The main highlights of our work are concluded as below.

1) Discriminant features of each single fault could be extracted from the compound fault signals without relying on expert knowledge and complicated signal processing.
2) Multi-label learning is combined with deep residual network to separately predict each single fault that composes a compound fault, thus diagnosing compound fault scientifically.
3) To reduce the dependence on labeled data, a multilayer domain adaptation module is designed to match the divergence, which enables the model to transfer the knowledge to the label-unavailable domain and adapt to the variable working conditions.

The rest of the paper is organized as follows. Section 2 introduces the basic idea and theoretical foundation of this work. Section 3 details the architecture and the concrete implementation of our method. Section 4 presents the experimental results and discussions. Finally, the conclusion for this paper is drawn in Section 5.

2. Ideas and theories

2.1. Multi-label learning for compound fault diagnosis

Multi-label learning aims to construct a classification model for objects that have multiple class attributes at the same time. As shown in Fig. 1, compared with the single-label sample which is only associated with a single class label, a multi-label sample could be connected with several different class labels [23]. Although each type of compound attribute can be treated as a single label [24], has proved that considering each single attribute in multi-label data is more consistent with the actual situation.

The present deep learning-based compound fault diagnosis methods tend to treat compound faults as an independent fault type, directly setting a new label to them for classification. Such methods need to exhaust all compound fault types, i.e., when there exist n single fault types, there will be 2^n-$(n+1)$ types of artificial compound faults. Obviously, the number of parameters and the training time will increase as the increase of fault types, which is prohibitive for model training. Furthermore, this treatment is not logically consistent with actual situation. For example, when facing an inner race fault, and a compound fault with outer race fault and inner race fault, the above methods treat them as mutually exclusive events. But they have similar features and probabilistic compatibility with each other. Therefore, the conventional multi-class classification is not applicable to the diagnosis of compound faults.

In multi-label learning, a sample could belong to multiple types at the same time, and different types are interrelated. That produces the prospect of avoiding the enumeration of compound fault types and considering the relationship between single fault and compound fault. The common techniques include Binary Relevance (BR), Label Combination, Ranking and Threshold method [25]. In this research, problem transformation in multi-label learning is used for compound fault diagnosis. It converts a multi-label problem into one or more single-label problems.

2.2. Domain adaptation for compound fault diagnosis

In practical engineering, due to the variable working condition, the labeled data of compound fault is always unavailable. Transfer learning is an effective approach to transfer the knowledge from the labeled training data to the unlabeled target data. Transfer learning applies knowledge learned from related tasks to solve the problems in different but relevant domains. In our method, since the labeled data is only available in source domain, a transductive transfer learning should be carried out based on the idea of domain adaptation.

Domain adaptation utilizes knowledge learned from related source domains to handle with the target tasks. Specifically, the domain can be expressed as $D = \{X\ ,\ P(X)\}$, where X represents the feature space, and $P(X)$ represents the marginal probability distribution of the sample. The task can be expressed as $T = \{Y\ ,\ f(\cdot)\}$, which includes the label space Y and the predictive function $f(\cdot)$. Knowledge could be transferred from task T_s based on source domain D_s to a target task T_t based on target domain D_t. Domain adaptation can significantly promote the precision of predictive function $f_t\ (\cdot)$ for the task T_t by reducing the distribution divergence between D_s and D_t in feature space.

In the fault diagnosis, the diversity of operating conditions will lead to difference in both marginal and conditional distributions of the samples in D_s and D_t, i.e., $\mathrm{P}(X_s) \neq \mathrm{P}(X_t)$, $P\ (Y_s \mid X_s) \neq P\ (Y_t \mid X_t)$. Domain adaptation could allow the model trained on the D_s to be applied to diverse but related D_t.

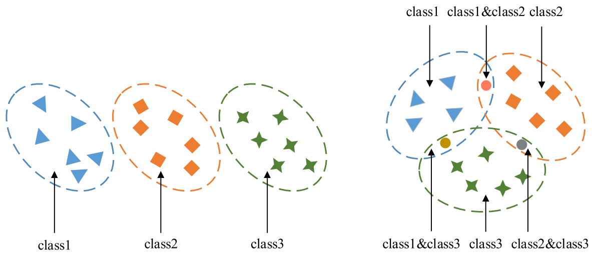


Fig. 1. (a) Single-label data; (b) multi-label data.

3. Proposed methods

3.1. Problem formulation

Compound faults are obviously distinct from single fault while also being related to them. Take the two bearings in Table 1 as an example. Fig. 2 shows the single fault and compound fault signals that collected from these two bearings at the same speed and load conditions. It is observed that compound fault signal has larger amplitude range than the single signal, and the peak magnitude at the characteristic fault frequency changes continuously under the coupling effect of multiple features. Due to the mutual interference between the different fault features, the compound fault signal normally does not have noticeable periodicity. And the single fault with weak characteristic signals becomes more inconspicuous under the influence of strong noise which make it difficult to be separated out.

Therefore, it is hard to extract the fault features directly from compound faults by the way of treating them as new fault type, independent of the single fault. A sensible approach is to separate the feature of each single fault from the compound faults and then classify them respectively. To achieve this, multi-label learning is introduced in this paper. Specifically, the problem transformation strategy in multi-label learning is used to transform the multi-label classification into binary classification. Thus, each type of single fault is dichotomized to determine its existence separately.

Besides, the lack of sufficient labeled data in target domain is also a common issue which results in the degradation of diagnosis model. For this problem, domain adaptation is adopted to map the data of D_s and D_t to a shared subspace and the Multiple Kernel-Maximum Mean Discrepancy (MK-MMD) is computed to match the marginal distribution differences. The proposed method is verified by experiments conducted under variable working conditions in Section 4.

3.2. The proposed multi-label domain adaptation method

In this paper, we propose a Multi-Label Domain Adaptive model (MLDA), and the architecture is shown in Fig. 3. The proposed MLDA consists of a fault feature extractor, a multi-label classifier, and a multilayer domain adaptation module.

1) MK-MMD

MMD is a typical approach for domain adaptation that measures the discrepancy by calculating the distance between the distributions in the RKHS [26], as follows:

$$\mathrm{MMD}^2(P,Q) = \|E_U[\varphi(\mathrm{x}^s)] - E_V[\varphi(\mathrm{x}^t)]\|^2_{\mathscr{H}_k} \tag{1}$$

where ★$_k$ is the RKHS, and P and Q represent two different distributions. RKHS is a space constructed by the kernel function k. For each RKHS has a unique corresponding k, while kernel function-based MMD could be formulated as follows:

$$\mathrm{MMD}_k^2(P,Q) = \frac{1}{{n_s}^2}\sum_{i,j=1}^{n_s} k\left(\mathrm{x}_i^s, \mathrm{x}_j^s\right) - \frac{2}{n_s n_t}\sum_{i,j=1}^{n_s,n_t} k\left(\mathrm{x}_i^s, \mathrm{x}_j^t\right) + \frac{1}{{n_t}^2}\sum_{i,j=1}^{n_t} k\left(\mathrm{x}_i^t, \mathrm{x}_j^t\right) \tag{2}$$

The feature mapping ability of RKHS is highly influenced by the choice of kernel function, which affects the effectiveness of MMD. However, the characterization capacity of a single kernel function is limited, which is well settled by MK-MMD. Gretton et al. [27] indicated that the MK-MMD method is able to minimize the type II error, i.e., the probability of incorrectly accepting the null hypothesis that the data distribution is the same, while it is actually different. Thus, the subspace mapping of multiple kernel functions is combined to get a more generalized feature representation.

In MK-MMD, the kernel function k is a linear combination of positive definite functions $\{k_u\}$, represented by the weighted sum of multiple kernels:

$$\Bbbk : = \left\{ k = \sum_{u=1}^{d} \beta_u k_u | \sum_{u=1}^{d} \beta_u = D > 0; \beta_u \geq 0, \forall u \right\} \tag{3}$$

Let ★$_k$ be the RKHS of functions $f: X \rightarrow R$ defined on X, and φ: $X \rightarrow$ ★$_k$ be the feature mapping that maps X onto ★$_k$. Then the MK-MMD between the source domain marginal distribution $\mathrm{P}(X_\mathrm{S})$ and target domain marginal distribution $\mathrm{P}(X_\mathrm{T})$ can be written as follows:

$$D_k^2(P(X_S), P(X_T)) = \left\|E_{P(X_S)}[\varphi(X_S)] - E_{P(X_T)}[\varphi(X_T)]\right\|^2_{\mathscr{H}_k} \tag{4}$$

Table 1
Description of tested bearing.

Bearing designation	Bearing fault pattern	Fault size
SKF6205-2RS	Inner race fault	0.2(mm)
SKF6205-2RS	Inner race fault & Ball fault	0.2(mm)

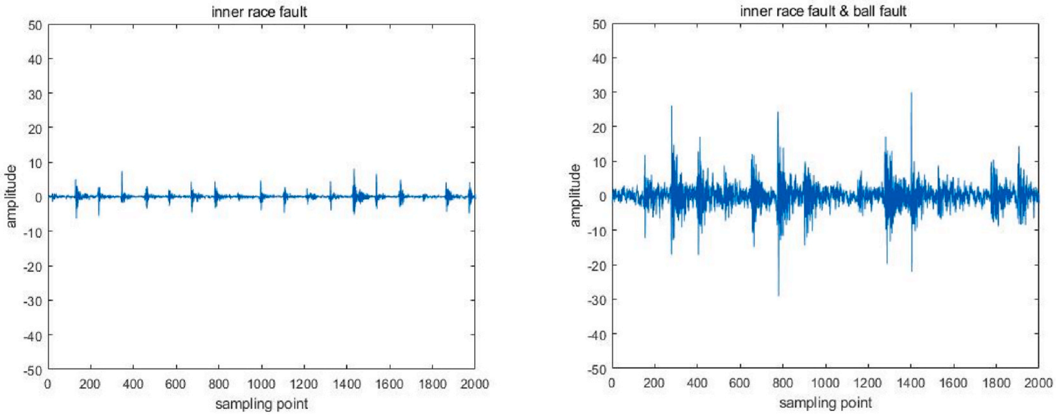


Fig. 2. (a) Inner race fault; (b) Inner race fault & Ball fault.

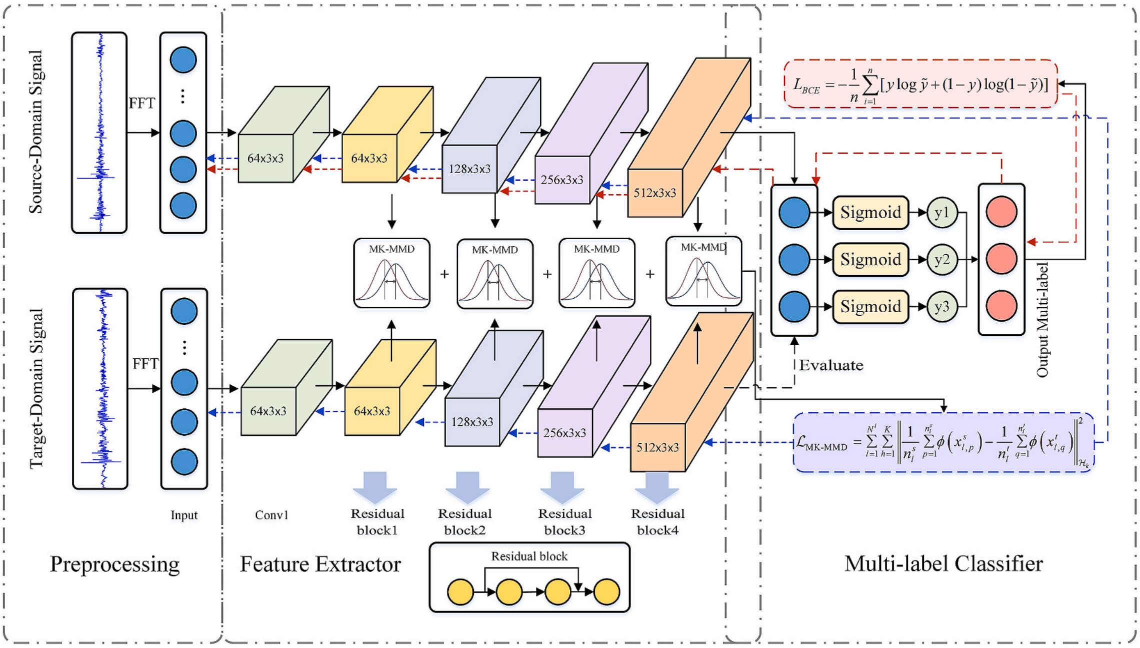


Fig. 3. The architecture of the proposed MLDA.

2) Multi-layer domain adaptation

As the depth of the network increases, the model tends to focus more on the overall deep features, but ignores some shallow information and local details. However, it will lead to the loss of information and impact the effectiveness of domain adaptation. To address this problem, multi-layer domain adaptation is employed in this article to match the distribution for the feature maps of D_s and D_t after each residual block. Based on the MK-MMD, the subspace mapping of multiple kernel functions is combined to extract features and match the marginal distribution in multiple layers of the network to obtain the domain-invariant features. After calculating the multilayer MK-MMD, the loss function of the distribution matching part could be written as:

$$\mathscr{L}_{\text{MK}-\text{MMD}} = \sum_{l=1}^{N^l} \sum_{h=1}^{K} \left\| \frac{1}{n_l^s} \sum_{p=1}^{n_l^s} \varphi\left(x_{l,p}^s\right) - \frac{1}{n_l^t} \sum_{q=1}^{n_l^t} \varphi\left(x_{l,q}^t\right) \right\|_{\mathscr{H}_k}^2 \tag{5}$$

where N^l denotes the number of layers for computing MK-MMD, K is the number of Gaussian kernels, $x_{l,p}^s$ and $x_{l,q}^t$ are the feature vectors of D_s and D_t extracted in the lth layer, n_l^s and n_l^t represent the number of samples in D_s and D_t.

3) Fault Feature Extractor

In this paper, a feature extractor with residual blocks is designed to get representative features. As an effective solution to the problem of vanishing gradient and exploding gradient, a deep residual neural network is commonly utilized in image recognition. Referring to the structure of ResNet, the feature extractor of the MLDA consist of an input layer, an activation layer, and four residual blocks. To accommodate the size of the input fault signal map, the kernel size of the convolutional layer in the network is set to 3×3. Denoting the learned residual function by F, the feature from the lth layer to the deeper Lth layer in the residual block is calculated as follow:

$$x_L = x_l + \sum_{i=1}^{L-1} F(x_i, W_i) \tag{6}$$

The residual block structure establishes a shortcut connection between the shallow and deep layers of the network, allowing a portion of the gradient calculation to be passed directly to the next layer in the form of summation, which becomes as follows:

$$\frac{\partial loss}{\partial x_l} = \frac{\partial loss}{\partial x_L} * \frac{\partial x_L}{\partial x_l} = \frac{\partial loss}{\partial x_L} * \left\{ 1 + \frac{\partial}{\partial x_L} \sum_{i=1}^{L-1} F(x_i, W_i) \right\} \tag{7}$$

Thus, the vanishing gradient and exploding gradient could be well settled, and the fault features of compound faults in shallow and deep layers are better preserved.

4) Multi-label classifier

The multi-label classifier designed in this paper transforms the task into multiple binary classification, with each binary classifier separately predicting a single label. Specifically, it consists of the following three steps：

1) Multi-label transformation: $(x,y) \rightarrow \{(x,y_j) | j = 1,...,C\}$,
2) Train multiple binary classifiers: $h = (h_1,...,h_C) : \chi \rightarrow \{0,1\}^C$,
3) Output the associated multi-label: $\widehat{y} = [y_1,...,y_C] = [h_1(x),...,h_C(x)]$,

Where C is the number of single fault type, $h : X \rightarrow Y$ is the mapping from sample input space to label output space, $\widehat{y} = h(x)$ is the multi-label prediction of sample x given by the classifier h.

For the compound faults of bearings consisting of three common forms of single fault, i.e., Inner Race Fault (IF), Outer Race Fault (OF) and Ball Fault (BF), C is taken as 3. The label of the ith fault can be expressed as a three-dimensional row vector $[y_i^1\ y_i^2\ y_i^3]$. Each fault category corresponds to one position of the row vector. For example, the inner race fault with ball fault can be represented as [1 0 1].

There are three neurons corresponding to three single faults in the output layer. Let the output of each neuron be s_i, and the Sigmoid function is used to transform it into the probability of the presence of this single fault, which is denoted:

$$P_i = \frac{1}{1 + e^{-s_i}} \tag{8}$$

Sigmoid can intuitively denote the activated states of the neuron, making it well suited for binary classification. A threshold function is used for the sigmoid activation detailed as follows:

$$F(x) = \begin{cases} 0 & 0 < x \leq \alpha \\ 1 & \alpha < x < 1 \end{cases} \tag{9}$$

where α is the threshold value for assessing the presence of each fault type. If the probability is greater than the threshold, the model would classify the sample as having this type of single fault. In this work, the value of α is set to 0.4, 0.45, 0.5, 0.55 and 0.6 respectively for the experiment. According to the contrast experiment, the model obtains the best performance when the threshold is set to 0.5. Therefore, the parameter α is fixed as 0.5 in this work. To minimize the classification error, Binary Cross-entropy(BCE) is utilized as the objective function, which is defined as follows：

$$L_{BCE} = -\frac{1}{n} \sum_{i=1}^{n} [y \log\widetilde{y} + (1-y)\log(1-\widetilde{y})] \tag{10}$$

Combining the three modules above, the total loss function of MLDA can be formulated as:

$$L_{total} = \lambda \sum_{l=1}^{N^l} \sum_{h=1}^{K} \left\| \frac{1}{n} \sum_{p=1}^{n} \varphi\left(x_{l,p}^s\right) - \frac{1}{n} \sum_{q=1}^{n} \varphi\left(x_{l,q}^t\right) \right\|_{\mathscr{H}_k}^2 - \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{C} \left[y_{i,j}^s \log h_j\left(x_i^s\right) + \left(1 - y_{i,j}^s\right) \log\left(1 - h_j\left(x_i^s\right)\right) \right] \tag{11}$$

where N^l is the number of layers for calculating MK-MMD, K represents the number of Gaussian kernels, λ is the tradeoff parameter, C denotes the number of single fault type, and $h_j(\cdot)$ is the mapping from the input samples to the jth label space. Adam is used to optimize the training process. Though the experiment, appropriate settings for the proposed problem are $lr = 0.0001, \beta_1 = 0.9, \beta_2 = 0.999$ and $\varepsilon = 10^{-8}$, where lr is the learning rate, β_1 and β_2 are the exponential decay rates for first and second order matrices, and ε is a constant added to maintain stability.

The procedure of compound fault diagnosis using MLDA is shown in Fig. 4, comprising the following steps.

1) Collect the vibration signals of rotating machinery under variable operating conditions. Convert the signal from time domain to frequency domain by Fast Fourier Transform (FFT), which makes the features of the fault more markedly. FFT transforms an intractable time-domain signal into a frequency-domain signal which is easy to analyze. The frequency-domain data after FFT is used as input to the model, which makes it easier to capture the useful features such as frequency, phase, and amplitude in the vibration signals.
2) Set the pre-processed frequency domain images to source domain and target domain respectively according to working conditions.
3) Pre-train the network with source domain data and then save the parameters for initializing.
4) Input the data of D_s and D_t into the pre-trained fault feature extractor, and measure difference of distribution for the features extracted by the residual block to get the transferable features.
5) Multi-label classifier provides a prediction of the existence of each single fault and output the combined multi-label.

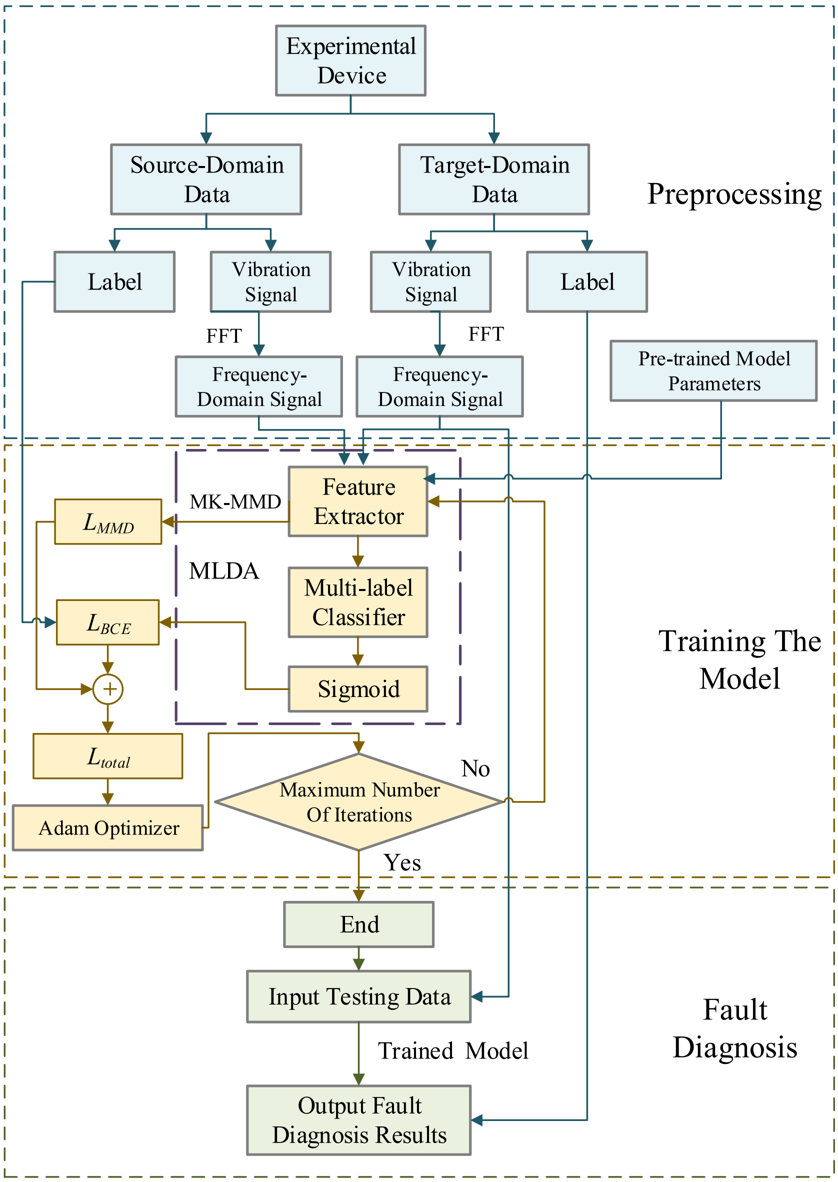


Fig. 4. Flowchart of the proposed MLDA.

6) Calculate the fault classifier loss based on the BCE, and combine it with the feature matching loss based on the multi-layer MK-MMD to get the total loss. Then the weights are updated with the error backward propagation and the Adam algorithm is used to optimize the network.
7) After the training is completed, the diagnostic performance of model is evaluated on target domain.

4. Experimental verification

4.1. Datasets description

To validate the effectiveness of the proposed MLDA, the bearing data acquired from the test rig of our lab are used. As shown in Fig. 5, the test rig consists of a drive motor, a plum coupling, a healthy bearing, an acceleration sensor, a loading device, and the testing bearing, while the bearing vibration signals are collected by the acceleration sensor with sampling rates of 10 kHz. The experimental bearing vibration signals captured under four different loads (0 kN, 1 kN, 2 kN, 3 kN) are set as four datasets C1, C2, C3, C4, and each dataset contains 320 training samples and 160 test samples. As shown in Table 2, for each dataset, there are eight different bearing health conditions, i.e., normal condition (NO), three single faults (IF, OF, BF) and four compound faults (IB, IO, OB, IOB), all with the fault size of 0.2 mm.

Based on the datasets C1、C2、C3 and C4, six different transfer tasks are set up for assessing the effect of each methods, which is shown in Table 3. For example, T1-2 denotes the transfer task using C1 as source domain and C2 as target domain.

4.2. Experimental results and performance analysis

4.2.1. Contrast experiments

In order to verify the effectiveness of MLDA, the comparative methods are conducted simultaneously, including ResNet, Transfer Component Analysis (TCA) [28], Joint Distribution Adaptation (JDA) [29] and Correlation Alignment (CORAL) [30]. ResNet is a classical deep learning method, and TCA, JDA, CORAL are common methods of transfer learning. All of comparative methods do not use multi-label classifier, but use One-Hot encoding and train according to traditional multi-classification. Among them, TCA learns a set of transformations to reduce the difference between the marginal probability distributions of D_s and D_t. JDA uses the source domain data to train a classifier and puts pseudo-labels on the target domain data. Then the pseudo-labels are used to calculate the conditional probability distribution of the target domain data. Minimize the difference between marginal distribution and conditional distribution at the same time, and achieve transfer between domains. CORAL is the transfer learning method based on CORAL-Loss, which consists of a feature extractor and a classifier. It is trained by classification loss of source domain and CORAL-Loss of features extracted from source and target domain data. The task settings of all experiments are training on labeled training data under one working condition to diagnose the unlabeled data under another working condition. The results of the comparison experiments are detailed in Table 4.

In order to intuitively compare the diagnostic effectiveness of each method on single fault and compound faults, the classification accuracy of these two kinds of bearing faults are counted respectively. As shown in Fig. 6, the proposed MLDA maintains the accuracy of 99% even when faced with the compound faults which are more difficult to identify. The results demonstrate the power of our method on compound fault diagnosis. By contrast, the traditional methods such as ResNet and JDA, show poor extraction capability for features of one or several types of faults. As a result, the models based on traditional methods could not recognize these types of faults, and the diagnostic performance fluctuate greatly.

The multi-label classifier in MLDA makes a great contribution to decompose compound fault features into corresponding single fault feature, and then accurate diagnosis could be achieved.

In terms of the adaptability to variable working condition, all other methods could not work well when faced with complex tasks such as T1-4 and T2-4 with remarkable changes in operating conditions, while the performance of MLDA model remains stable. The result indicates that MLDA has a superior ability to learn domain invariant features. The multi-layer domain adaptation could capture the fault features in multiple dimensions for D_s and D_t, thus effectively reducing the distribution discrepancy between them and

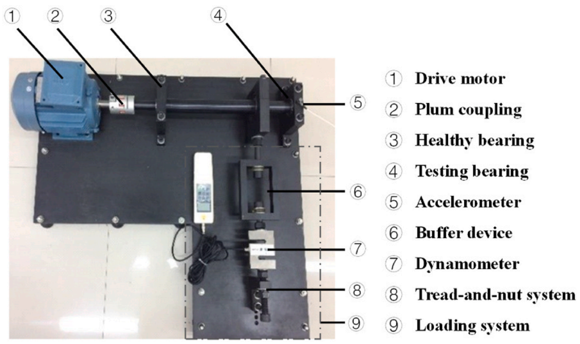


Fig. 5. The bearing test rig.

Table 2
Description of bearing datasets.

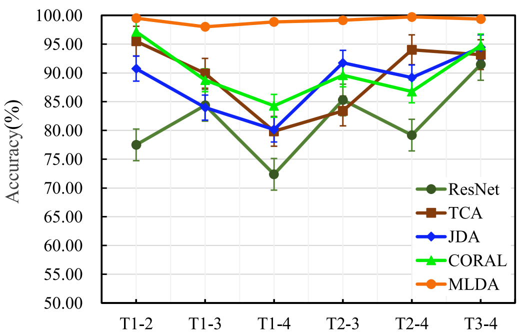
Bearing fault pattern	Diameters	Label	Abbreviation
Normal	–	[0 0 0]	NO
Inner race fault	0.2(mm)	[1 0 0]	IF
Ball fault	0.2(mm)	[0 1 0]	BF
Outer race fault	0.2(mm)	[0 0 1]	OF
Inner race with Ball fault	0.2(mm)	[1 1 0]	IB
Inner race with Outer race fault	0.2(mm)	[1 0 1]	IO
Outer race with Ball fault	0.2(mm)	[0 1 1]	OB
Inner with Outer and Roller fault	0.2(mm)	[1 1 1]	IOB

Table 3
Transfer task setting.

Task	Source domain	Target domain
T1-2	C1 (0 kN)	C2 (1 kN)
T1-3	C1 (0 kN)	C3 (2 kN)
T1-4	C1 (0 kN)	C4 (3 kN)
T2-3	C2 (1 kN)	C3 (2 kN)
T2-4	C2 (1 kN)	C4 (3 kN)
T3-4	C3 (2 kN)	C4 (3 kN)

Table 4
The compared results of different models.

Task	Diagnosis accuracy for single fault (%)					Diagnosis accuracy for compound fault (%)				
	ResNet	TCA	JDA	CORAL	MLDA	ResNet	TCA	JDA	CORAL	MLDA
T1-2	77.50	95.51	90.76	97.16	99.52	70.47	94.25	93.72	95.16	99.49
T1-3	84.37	89.94	83.97	88.73	98.02	73.07	83.19	91.44	84.28	99.64
T1-4	72.39	79.86	80.15	84.29	98.89	69.84	72.08	82.07	80.06	97.33
T2-3	85.31	83.39	91.74	89.58	99.14	76.81	92.66	96.87	92.33	99.05
T2-4	79.04	94.03	89.2	86.78	99.74	89.69	85.71	80.94	83.17	99.32
T3-4	91.48	93.18	94.44	94.78	99.38	83.47	90.36	94.75	92.86	99.42
Average	81.68	89.32	88.38	90.22	99.12	77.23	86.38	89.97	87.98	99.04



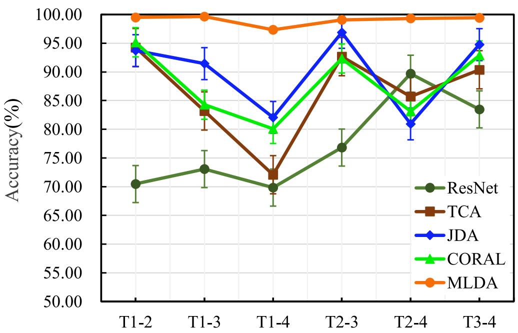


Fig. 6. (a) Classification accuracy for single fault; (b) Classification accuracy for compound fault.

improving the robustness of the model.

In the proposed MLDA, the tradeoff parameter λ is important to affect the performance, which balances the training of the multi-label classifier and the distribution adaptation. Thus, we analyze the performance of MLDA under different tradeoff parameters. During the experiments, the parameter λ is chosen from {0.01, 0.1, 0.5, 1, 1.5, 5, 10}. Fig. 7 presents the final loss under different tradeoff parameters. As shown in Fig. 7, when the λ is small, the loss decreases with the increase of λ, indicating that too small balance parameters may make the model unable to be fully trained. However, if λ continues to increase, the loss will also increase, indicating that excessive value of tradeoff parameter will affect the convergence of loss functions of the classifier. According to the above analyses and the experiment, the parameter λ is set as 1.5 in this case.

To further compare the diagnostic effectiveness of various methods, Fig. 8 shows the diagnostic results of three different methods for a sample with IOB fault under the T2-3 transfer task. After visualizing the features extracted by *t*-SNE, the proposed MLDA

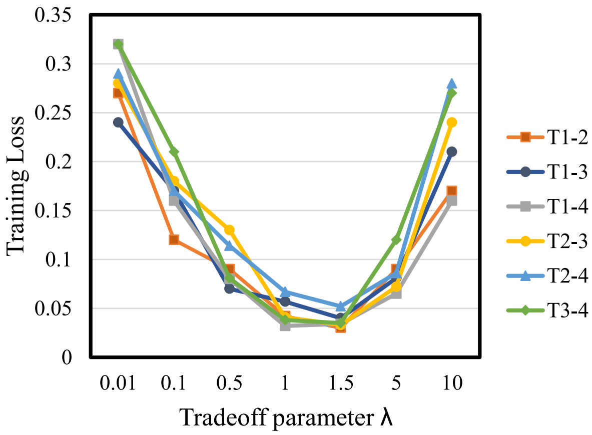


Fig. 7. Total loss under different tradeoff parameter.

effectively reduced the distance between the D_s and D_t, compared to traditional methods such as TCA and CORAL. From the results, for the task T2-3 with relatively lower difficulty, TCA can diagnose correctly, while CORAL misdiagnosed it as OB fault. And the MLDA achieved successful prediction with the output probability of [0.96 0.99 0.98], which is regarded as a label [1 1 1] according to the threshold function of sigmoid.

However, in a more complex transfer task T1-4, Fig. 9 shows that both TCA and CORAL misjudged a sample with OB fault. While the MLDA model gave a probability of [0.11 0.88 0.98], and then it was transformed into [0 1 1] by the threshold function, which was the correct prediction.

4.2.2. *Ablation experiment*

To specifically investigate the effectiveness of the multi-label classifier, MK-MMD, and multi-layer domain adaptive methods used in the MLDA, a series of ablation experiments are conducted in this section.

1) To validate the effectiveness of the multi-label learning strategy for compound fault diagnosis, an experiment is conducted by replacing the multi-label classifier in the model with a Softmax classifier, represented as MLDA#1.
2) To analyze the effect of MK-MMD, we replace it with the MMD with only one Gaussian kernel, represented as MLDA#2.
3) To validate the effectiveness of the multi-layer domain adaptation, MK-MMD is calculated only for the features extracted from a single residual block in the experiment named MLDA#3.

The experimental results are shown in Fig. 10.

According to the results of the ablation experiments shown in Fig. 10, the proposed MLDA outperforms other ablation methods, with an average accuracy of 99.36%. It shows that the performance of MLDA#1, with the multi-label classifier replaced, is heavily influenced by the variable load. It performs well for the transfer tasks of T2-3, T2-4 and T3-4. However, when faced with the tasks transferring from the unloaded working condition to loaded condition, the performance of the model significantly degenerates. The results indicate that it is difficult to learn the essential features of compound faults by regarding them directly as integrated patterns. That's why the model could not work well when there was remarkable difference between D_s and D_t. Hence, we can have the conclusion that the multi-label classifier contributes to learn the discriminant features of each single fault and improve the

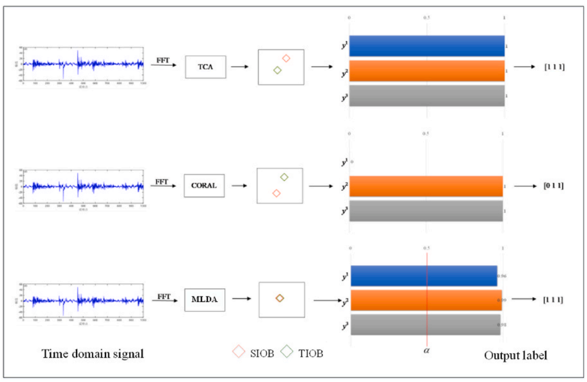


Fig. 8. The diagnosis results of an IOB fault by different methods.

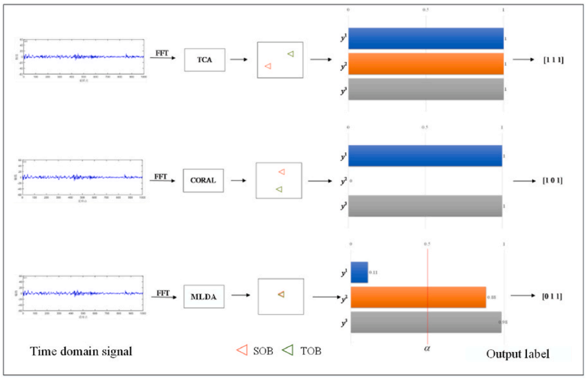
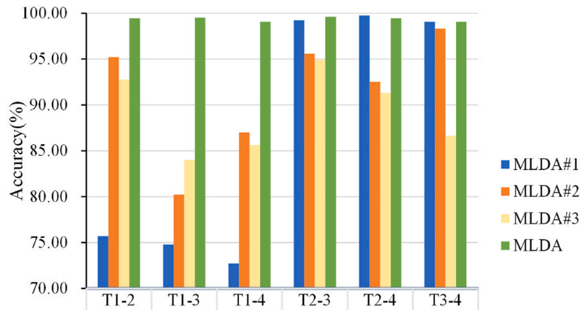


Fig. 9. The diagnosis results of an OB fault by different methods.



	T1-2	T1-3	T1-4	T2-3	T2-4	T3-4
MLDA#1	75.70	74.77	72.73	99.22	99.77	99.06
MLDA#2	95.21	80.21	86.99	95.57	92.53	98.32
MLDA#3	92.76	84.00	85.63	94.91	91.34	86.64
MLDA	99.45	99.53	99.06	99.61	99.45	99.06

Fig. 10. Diagnosis results of different experiments.

interpretability of the model.

The diagnosis accuracies of MLDA#2 and MLDA#3 are relatively lower, and both could not adapt the variation of working condition. The above results demonstrate the necessity of MK-MMD and multi-layer domain adaptation. Actually, if the feature matching is only done for a single layer, some useful information and local details would be overlooked, leading to domain bias. With the multi-layer domain adaptation, MLDA limits the discrepancy for both shallow and deep representation to get the transferable features so that it achieves great performance. In a word, the effectiveness and superiority of the methods used in the proposed MLDA have been validated well by the ablation experiments.

4.2.3. *Feature visualization*

In order to further analyze whether the proposed MLDA extracts the transferable features, *t*-SNE is introduced to visualize the features extracted by TCA, CORAL, MLDA#3, and MLDA. Fig. 11 shows the experimental results under the T1-4 task, which indicates that TCA and CORAL have poor effects in clustering. Both of them have difficulty in reducing the distance between the source and target domains when the work condition is rather variable. The MLDA#3, without multi-layer domain adaptation, only aligns the statistical data center of the features although the clustering effect is better. It can be observed that MLDA significantly improves the detachable degree of compound faults and thus discriminates the examples more precisely. Moreover, the representations of D_s and D_t are nearly projected to the same region with MLDA. This result proves that our method could match the distributions discrepancy between source and target domains and enhance the similarity of the extracted features from them. Thus, it could learn the domain-invariant features to achieve an accurate diagnosis of the compound faults in unlabeled target domains.

5. Conclusions

Aiming to diagnose the compound fault of rolling bearing under variable working conditions, a novel diagnosis method is presented in this paper. The case studies show that the multi-label classifier contributes to learn the discriminant features of each single fault and improve the interpretability of the model. By multi-layer domain adaptation with multi-kernel embedding matching, the distribution of source domain and target domain is effectively aligned, thus effectively resolving the problem of domain-shift. Experiments on the bearing dataset prove that this method significantly improves the similarity of transferrable features and discriminates the examples

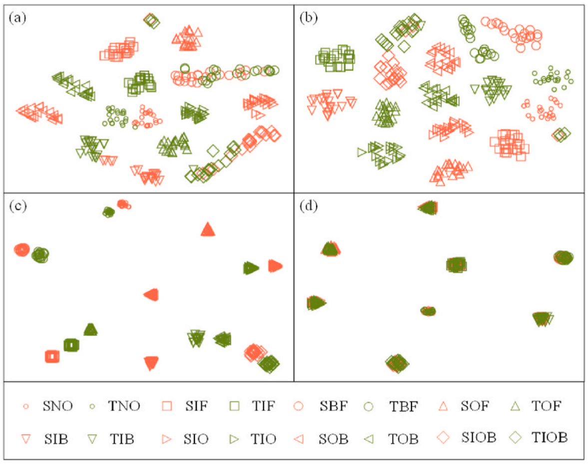


Fig. 11. Feature visualization for task T1-4
(a) TCA; (b) CORAL; (c) MLDA#3; (d) MLDA.

more precisely. So that it still maintains a stable diagnostic performance when faced with variable working conditions. The combination of multi-label learning and deep domain adaptation could extract the transferable features of compound faults, and reach the accurate diagnosis even facing unlabeled data in target domain.

In future work, the proposed method will be tested for industrial application and will attempt to diagnose unseen compound fault types. Furthermore, unsupervised learning and data augmentation methods will be attempted to combine with the current method. In this way, it is possible to accurately diagnose compound faults while only applying normal and single fault data for training. Moreover, the advanced compound fault expression can be tested to further identify compound faults with different damage degrees.

Author contribution statement

Liuxing Chu: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Wrote the paper. Qi Li, Changqing Shen, Dong Wang: Analyzed and interpreted the data; Wrote the paper. Bingru Yang, Liang Chen: Conceived and designed the experiments; Wrote the paper. Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Contributed reagents, materials, analysis tools or data; Wrote the paper.

Funding statement

Prof. Liang Chen was supported by National Innovation and Development Project of Industrial Internet [TC190H3WR].

Changqing Shen was supported by National Natural Science Foundation of China [51,875,375], China Postdoctoral Science Foundation funded project [2021M701503].

Data availability statement

The authors do not have permission to share data.

Declaration of interest's statement

The authors declare no competing interests.

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